SOLUTIONS: DISCRETE EVENT SYSTEMS
MASTER IN CONTROL AND OPTIMISATION

1. Exercise

   a) We denote states as $X = \{-1, 0, 1\}$. In particular $x \in X$ is the value of the difference of pieces of type 1 and 2 respectively. The event set is $E = \{a_1, a_2\}$ and the associated transition diagram is shown in Fig. 1.1. The initial state is 0. [4]

   b) The extended automaton G_F is shown in Fig. 1.2. [4]

   c) The labelling automaton has two states, $X = \{N, Y\}$, initial state is N, and two transitions, namely $f_L(N, f) = Y$ and $f_L(Y, f) = Y$. [2]

   d) The parallel composition $G_F || G_L$ is shown in Fig. 1.3. [3]

   e) The observer automaton $\text{Obs}(\tilde{G}_N)$ is shown in Fig. 1.4. [5]

   f) The state reached after the word $w = a_2a_1a_1a_2a_2a_2a_1a_1$ is detected can be seen from the transition diagram to be $\{1Y, 2Y\}$ in the Observer automaton. This means that the fault event has occurred (due to the Y label being present in all states) and buffer B_1 has 1 or 2 pieces more than buffer B_2. [2]

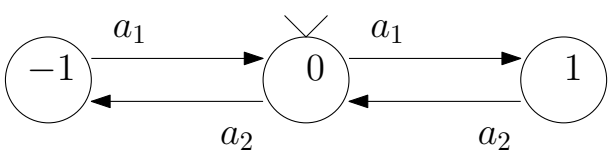


Figure 1.1 Automaton G, modelling the relative occupancy of buffers B_1 and B_2

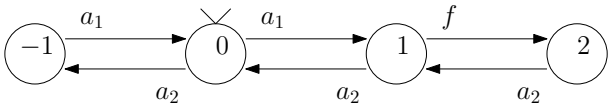


Figure 1.2 Automaton G_F

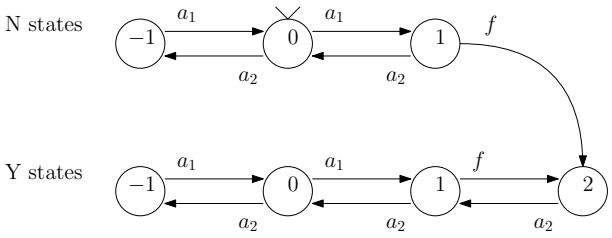


Figure 1.3 Automaton $G_F||G_L$

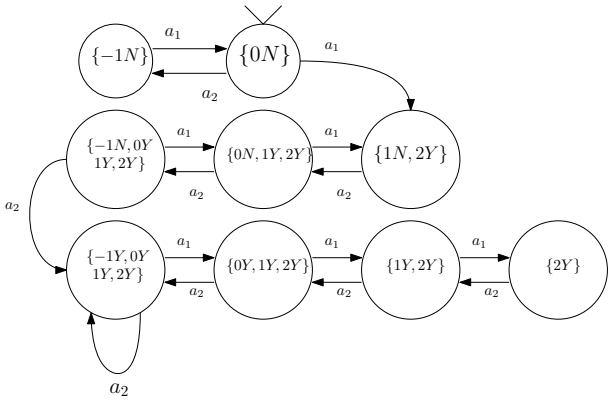


Figure 1.4 Observer automaton Obs$(\tilde{G}_N)$

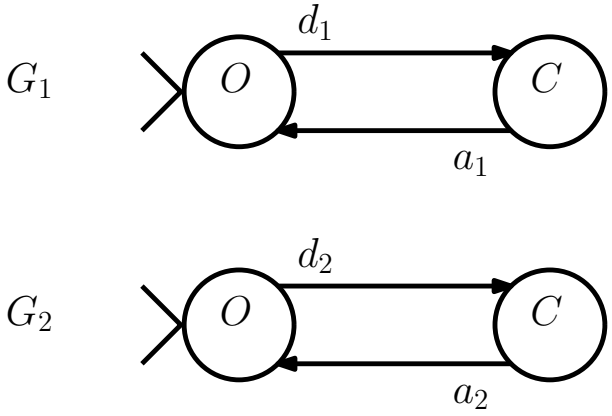


Figure 2.1 Automatoa G_1 and G_2

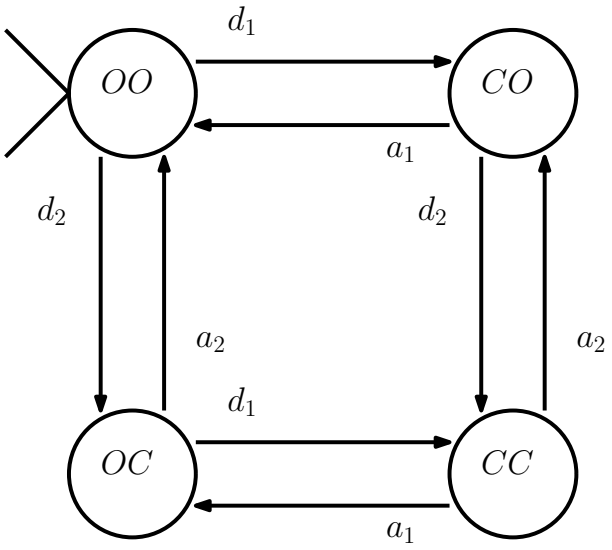


Figure 2.2 Automaton $G = G_1||G_2$

2. Exercise

a) The Automata modelling the individual robots and associated docking and activation events are shown in Fig. 2.1. [3]

b) The parallel composition $G = G_1||G_2$ is shown in Fig. 2.2. [3]

c) The FIFO policy specification can be implemented through state splitting as shown in Fig. 2.3. [4]

d) To find out whether the specification is controllable, given a set of uncontrollable events E_{uc} we compute the parallel composition $G||G_K$ and, for each state (x,x_K) of the composition we compare the enabled events with those enabled in x. The parallel composition $G||G_K$ is shown in Fig. 2.4. To emphasize the states where $\Gamma((x,x_K))\backslash\Gamma(x)$ is non-empty we have used dashed arrows. Notice that in state $CC,CC1$ event a_1 is disabled and in state $CC,CC2$ event a_2 is disabled. Since neither is an uncontrollable event when $E_{uc} = \{d_1,d_2\}$, the specification is controllable. [3]

e) When $E_{uc} = \{a_1\}$ the original specification is uncontrollable as a_1 is disabled

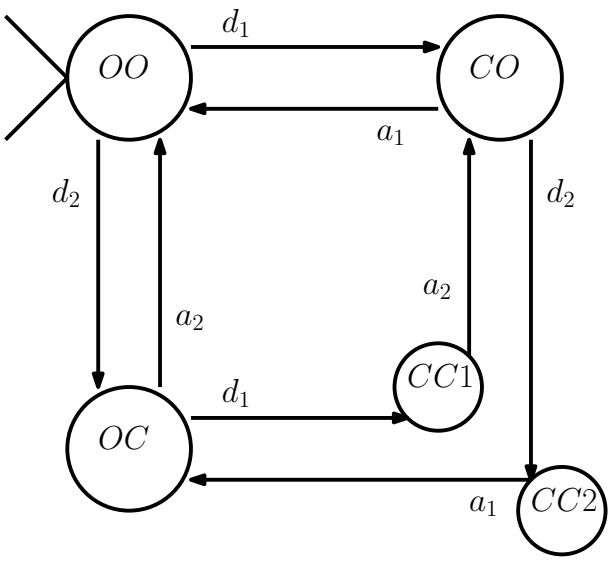


Figure 2.3 Specification Automaton G_K

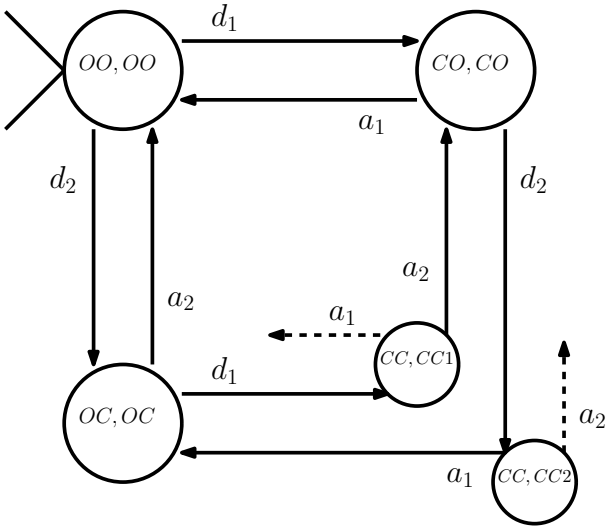


Figure 2.4 Parallel composition $G||G_K$

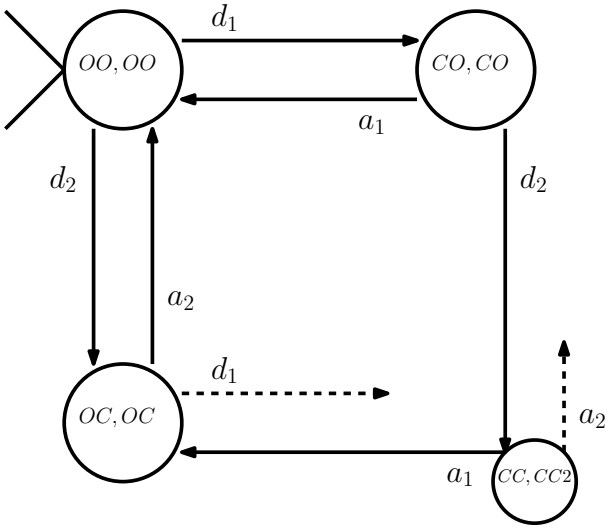


Figure 2.5 Supervisor Automaton implementing $\mathscr{L}(G_K)^{\uparrow C}$

in state $CC, CC1$. Therefore this state is removed, leading to the automaton in Fig. 2.5. This admissible supervisor implements the maximal controllable sublanguage. [5]

f) There are $3! = 6$ possible permutations in which 3 robots can enter the charging area. These will need to be accounted for in the specification automaton, leading to a state splitting specification where state *CCC* appears as 6 distinct states. [2]

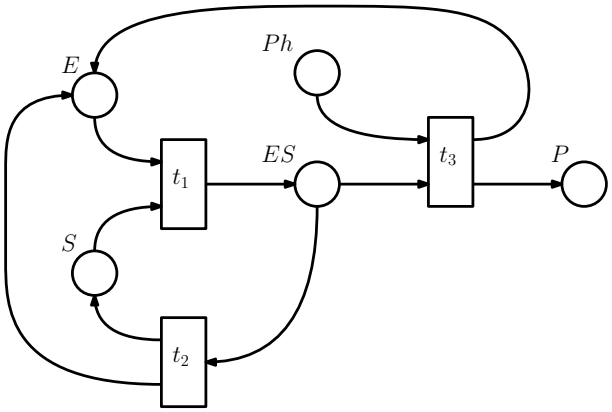


Figure 3.1 Petri Net N

3. Exercise

a) We have the following biochemical reactions: $E+S \to ES$, $ES \to E+S$, $Ph+ES \to E+P$. Accordingly we adopt a Petri Net with 5 places $\{E,S,ES,P,Ph\}$ and 3 transitions, as shown in Fig. 3.1. [6]

b) Ordering places as listed above, the incidence (stoichiometry) matrix C reads:

$$C = \begin{bmatrix} -1 & 1 & 1 \\ -1 & 1 & 0 \\ 1 & -1 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & -1 \end{bmatrix}.$$

[2]

c) The P semiflows correspond to the conservation laws of the checmial reaction. Namely, conservation of Substrate $S+ES+P$, Enzyme $E+ES$ and Phosphate $Ph+P$, respectively. These are the minimal P-semiflows: $v_1 = [0,1,1,1,0]$, $v_2 = [1,0,1,0,0]$ and $v_3 = [0,0,0,1,1]$. [4]

d) The incidence matrix for the modified Petri Net $\tilde{N}$ (see Fig. 3.2) equals:

$$\tilde{C} = \begin{bmatrix} -1 & 1 & 1 & 0 \\ -1 & 1 & 0 & 1 \\ 1 & -1 & -1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & -1 & 0 \end{bmatrix}.$$

Notice that $v_3\tilde{C} = [0,0,0,-1] \leq 0$. Hence v_3 is a P-decreasing vector and not a P-semiflow. Notice that conservation of Ph is compromised as in the degradation phase there is no release of the Ph molecule, but only a transformation of P into S. [3]

e) The network N is weakly conservative as $v_1+v_2+v_3$ is a P-semiflow that 'covers' all places. On the other hand, no such P-semiflow exists for $\tilde{N}$ which is therefore not structurally conservative. [3]

f) Being weakly conservative also implies structural boundedness of N. On the other hand, $v_1+v_2+v_3$ is a P-decreasing vector for $\tilde{N}$ which covers all places. Therefore $\tilde{N}$ is also structurally bounded. [2]

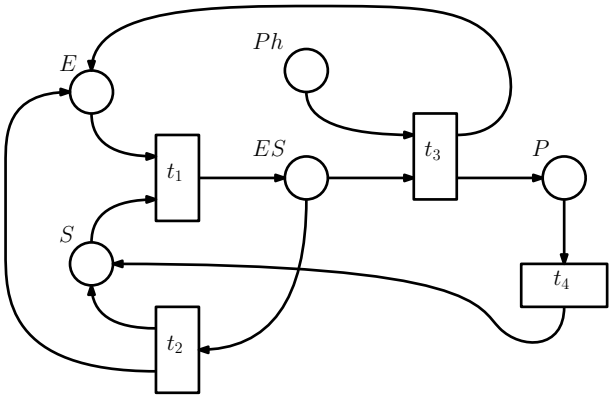


Figure 3.2 Petri Net $\tilde{N}$

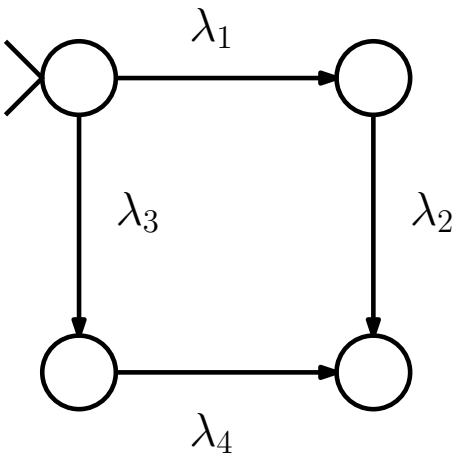


Figure 4.1 Markov chain modelling the machine *M*

4. Exercise

a) The transition diagram of the Markov chain is shown in Fig. 4.1 [5]

b) The vector π evolves according to

$$\dot{\pi} = \pi Q$$

with:

$$Q = \begin{bmatrix} -(\lambda_1+\lambda_3) & \lambda_1 & \lambda_3 & 0 \\ 0 & -\lambda_2 & 0 & \lambda_2 \\ 0 & 0 & -\lambda_4 & \lambda_4 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

[3]

c) The average time it takes to complete a toy house is the expected value of the absorption time τ_a of state 4, from initial state 1. Its probability density function is therefore $\dot{\pi}_4(t)$. We can access this function in the *s* domain through the formula:

$$\mathscr{L}(\dot{\pi}_4) = [1,0,0] \begin{bmatrix} s+\lambda_1+\lambda_3 & -\lambda_1 & -\lambda_3 \\ 0 & s+\lambda_2 & 0 \\ 0 & 0 & s+\lambda_4 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ \lambda_2 \\ \lambda_4 \end{bmatrix}$$

$$= \frac{(\lambda_1\lambda_2+\lambda_3\lambda_4)s+\lambda_1\lambda_2\lambda_4+\lambda_2\lambda_3\lambda_4}{(s+\lambda_2)(s+\lambda_4)(s+\lambda_1+\lambda_3)}$$

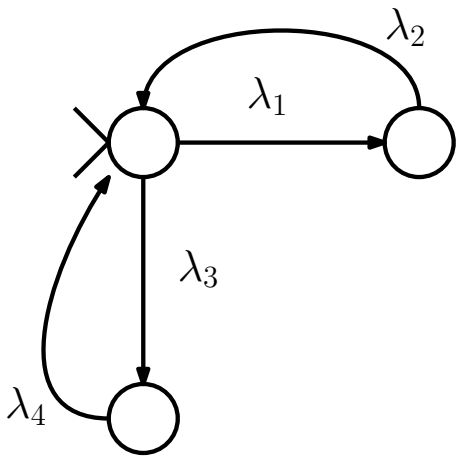


Figure 4.2 Modified Markov chain modelling the machine *M*

The average absortion time is computed as:

$$E[\tau_a] = \int_0^{+\infty} t\dot{\pi}_4(t)\,dt = -\frac{d}{ds}\mathscr{L}(\dot{\pi}_4)\Big|_{s=0} = \frac{\lambda_1\lambda_4 + \lambda_2\lambda_3 + \lambda_2\lambda_4}{\lambda_2\lambda_4(\lambda_1+\lambda_3)}.$$

[5]

d) We modify the Markov chain according to the diagram in Fig. 4.2. The corresponding equations read as $\dot{\pi} = \pi\tilde{Q}$ with $\tilde{Q}$ given as:

$$\tilde{Q} = \begin{bmatrix} -(\lambda_1+\lambda_3) & \lambda_1 & \lambda_3 \\ \lambda_2 & -\lambda_2 & 0 \\ \lambda_4 & 0 & -\lambda_4 \end{bmatrix}.$$

[2]

e) To find the fraction of time in the idle state we compute the steady-state probability distribution, by solving the equations:

$$\pi\tilde{Q} = 0, \qquad \pi_1 + \pi_2 + \pi_3 = 1$$

This gives:

$$\pi_\infty = [\lambda_2\lambda_4, \lambda_1\lambda_4, \lambda_2\lambda_3]/(\lambda_2\lambda_4 + \lambda_1\lambda_4 + \lambda_2\lambda_3).$$

Specifically, the fraction of time spent in the idle state is given by:

$$\pi_{\infty 1} = \frac{\lambda_2\lambda_4}{\lambda_2\lambda_4 + \lambda_1\lambda_4 + \lambda_2\lambda_3}.$$

[3]

f) Notice that the modified Markov chain is ergodic, (its transition diagram is in fact irreducible). According to the Ergodic Theorem, with probability 1 realizations $x(\cdot)$ of the chain will spend $\pi_{\infty 1}$ of their time in state 1 (idle state).

[2]